National University of Computer and Emerging Sciences, Lahore Campus

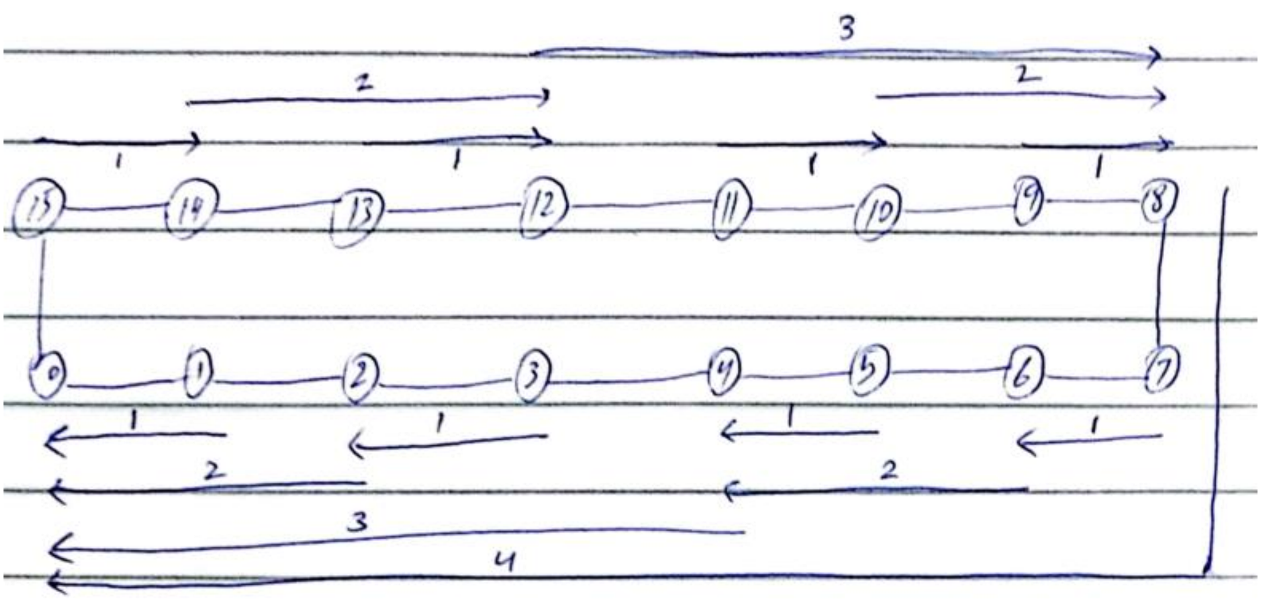
	Course:	Parallel and Distributed Computing	Course Code:	CS-3006
	Program:	BSCS & BSDS	Semester:	Spring 2024
	Duration:	20 Minutes	Total Marks:	14
	Paper Date:	02-Apr-2024	Weight	3.34%
	Type:	Announced	Page(s):	2
	Exam:	Quiz 3B Solution	Section:	

Name & Roll No:

Attempt all questions on the question booklet. Rough sheets can be used but it should not be attached. State answers in readable handwriting and with the help of diagrams, where necessary.

Question # 1: **[2 + 4 marks, CLO # 3]**

Draw 16 nodes ring structure. Considering message on all nodes, perform all-to-one reduction on 16 nodes ring structure. Node 0 is the destination node of all-to-one reduction operation. Show all the steps involved in the all-to-one reduction operation.

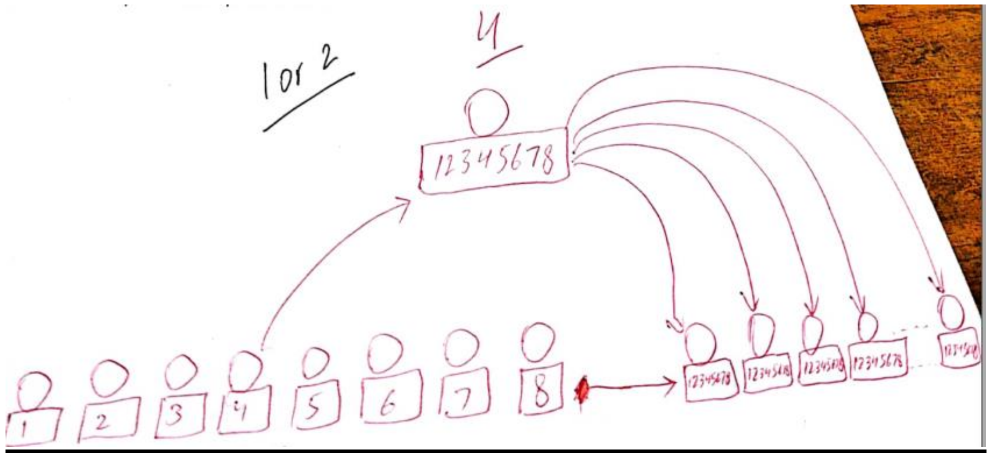


Question # 2: **[1 + 3 marks, CLO # 3]**

Define all-reduce operation. Perform All reduce operation on 8 nodes structure.

All-Reduce:

- Use all-to-one reduction followed by one-to-all broadcast.



Question # 3: **[2 + 2 marks, CLO # 3]**

Explain the difference between one-to-all broadcast and scatter operation with the help of diagrams.

One-to-All Broadcast:

- **A single process sends identical data to all other processes.**
    - **Initially one process has data of m size.**
    - **After broadcast operation, each of the processes have own copy of the m size.**

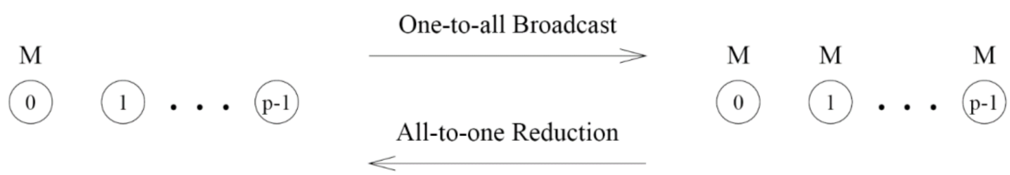


Figure 4.1 One-to-all broadcast and all-to-one reduction.

Scatter Operation:

- **A single process sends different data to all other processes.**

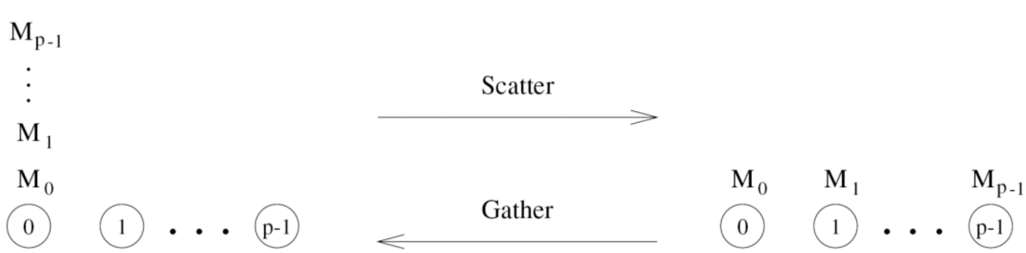


Figure 4.14 Scatter and gather operations.